

AMENDMENT NO. 1 MARCH 2023

TO

**IS 4985 : 2021 UNPLASTICIZED PVC PIPES FOR WATER
SUPPLIES — SPECIFICATION**

(Fourth Revision)

(Page 4, Clause 7.1.2, Para 2, line 2) — Substitute ‘5.1’ for ‘2.1.1 and 2.1.2’.

(Page 11, Clause 8, line 1) — Substitute ‘classes’ for ‘clauses’.

(Page 12, Clauses 10.3 and 10.3.1) — Substitute the following for the existing clauses:

‘10.3 Effect on Water

The pipes shall not have any detrimental effect on the composition of water flowing through them. When tested by the method described in IS 12235 (Part 4) and IS 12235 (Part 10), the quantities of lead, dialkyl tin C4 and higher homologues (measured as tin), cadmium, mercury and any other toxic substances extracted from the internal walls of the pipes shall not exceed the following concentrations in the test solution:

a)	Lead (first extraction)	:	1.0 mg/l (1.0 ppm by mass)
b)	Lead (third extraction)	:	0.05 mg/l (0.05 ppm by mass)
c)	Dialkyl tin C4 and higher homologues (measured as tin) (third extraction)	:	0.02 mg/l (0.02 ppm by mass)
d)	Cadmium (mean of first extraction, second extraction and third extraction)	:	0.01 mg/l (0.01 ppm by mass)
e)	Mercury (mean of first extraction, second extraction and third extraction)	:	0.001 mg/l (0.001 ppm by mass)
f)	Other toxic substances such as ‘di-n-octyl-tin-s-s bis iso-octyl mercapto acetate’ and ‘butyl stearate’ (third extraction)	:	0.01 mg/l (0.01 ppm by mass)

Price Group 1

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However, the pipes manufactured without lead stabilizer, which are also termed as Lead-Free (LF) shall meet the requirements of lead extraction as given below:

Lead (first extraction)	:	0.01 mg/l (0.01 ppm by mass)
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The limits for dialkyl tin C4 and higher homologues (measured as tin), and any other toxic substances shall remain the same as given in **10.3** (c) to (f). These pipes shall be marked as LF (*see 13.1*).

10.3.1 The manufacturer, for the purpose of these tests, shall disclose any other toxic substances present. The limit of all the 'other toxic substances', including the two mentioned in **10.3** shall not exceed the specified limit of 0.01 mg/l.

NOTE — Implementation of the lead phase out programme of the Government of India for phasing out use of lead stabilizers in PVC pipes and fittings manufacturing, shall be borne in mind'.

[Page 13, *Clause 13.1* (e)] — Insert the following after SI No. (e):

'f) The symbol LF in case of lead-free pipes'.

(Page 15, *Title of IS 4669 : 1968*) — Substitute 'Polyvinyl' for 'Polyrinyl'.

(Page 23, *Clause D-1.4.1*) — Substitute the following for the existing clause:

'The number of test samples to be taken from a lot shall depend on the size of the lot and shall be in accordance with Table 13'.

(Page 23, *Clause D-1.4.3, last line*) — Substitute 'col 6' for 'col 5'.

(Page 23, *Clause D-1.5.2, line 6*) — Substitute 'standard' for 'specification'.

[Page 24, *Table 13, SI No. (iii), col (4)*] — Substitute '32' for 's32'.

(Page 24, *Clause D-1.6.1, line 3*) — Delete 'as given in Table 14'.

(Page 24, *Clause D-1.8.2*) — Substitute the following for the existing clause:

'For this purpose, the procedure adopted for sampling and criteria for conformity shall be the same as that for reversion test under **D-1.5.2**, using Table 15'.

[Page 25, *Table 15, SI No. (i), col (7)*] — Substitute '2' for '1'.

(Page 25, *Clause D-1.9.2*) — Substitute the following for the existing clause:

'For this purpose, the procedure adopted for sampling and criteria for conformity shall be the same as that for reversion test under **D-1.5.2**, using Table 16'.

(Third cover page, *Foreword, last para, line 3*) — Substitute 'IS 2 : 2022 'Rules for rounding off numerical values (second revision)' for 'IS 2 : 1960 'Rules for rounding off numerical values (revised)'.

(CED 50)